BSDS V: Regression Techniques

Problems Set 2: Least Squares Regression

Throughout this problem set, we will consider the multiple linear regression model with response variable y and a list of regressors/covariates $\{x_1, \ldots, x_p\}$. Based on n observations the model is

$$\underset{\sim}{Y} = X\underset{\sim}{\beta} + \underset{\sim}{\varepsilon}, \quad \text{where} \quad \underset{\sim}{Y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}, \quad X = \begin{bmatrix} \underset{\sim}{1} & \underset{\sim}{x}_{(1)} & \ldots & \underset{\sim}{x}_{(p)} \end{bmatrix}, \quad \underset{\sim}{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_p \end{bmatrix}, \quad \text{and} \quad \underset{\sim}{\varepsilon} = \begin{bmatrix} \varepsilon_1 \\ \vdots \\ \varepsilon_n \end{bmatrix}. \tag{1}$$

We assume that $E(\underset{\sim}{\varepsilon}) = \underset{\sim}{0}$ and $var(\underset{\sim}{\varepsilon}) = \sigma^2 I_n$.

A. Problems without any distributional assumption on $\underset{\sim}{\varepsilon}$.

1. Consider the least square solution of $\underset{\sim}{\beta}$, say $\widehat{\underset{\sim}{\beta}}_{\text{OLS}}$, obtained from the linear regression model in (1). Further, consider the j-th simple linear regression model

   $$Y_i = \beta_0 + \beta_j x_{i,j} + \varepsilon_i, \quad i = 1, \ldots, n,$$

   and let $\widehat{\beta}_{j,S}$ be the solution to the simple linear regression model, and $\widehat{\beta}_S = \begin{bmatrix} \bar{Y} & \widehat{\beta}_{1,S} & \ldots & \widehat{\beta}_{p,S} \end{bmatrix}^\top$.

   (a) Compare $\widehat{\underset{\sim}{\beta}}_{\text{OLS}}$ and $\widehat{\beta}_S$.

   (b) When will these two solutions be equal?

2. Consider the regression model in (1), and let the design matrix satisfy the following orthogonality property:

   $$\underset{\sim}{1}^\top \underset{\sim}{x}_{(j)} = 0, \quad \text{and} \quad \underset{\sim}{x}_{(k)}^\top \underset{\sim}{x}_{(j)} = 0, \quad \text{for all } k \neq j, \quad j, k = 1, \ldots, p.$$

   (a) Find the expression of $\widehat{\underset{\sim}{\beta}}_{\text{OLS}}$ under this situation.

   (b) Find the expression of the residual sum of squares (RSS) under this situation.

3. Consider the regression model in (1), and let the design matrix satisfy the following orthogonality property:

   $$\underset{\sim}{1}^\top \underset{\sim}{x}_{(j)} = 0, \quad \text{and} \quad \|\underset{\sim}{x}_{(j)}\| = c, \quad \text{for all } j = 1, \ldots, p.$$

   Show that $\sum_{j=1}^{p} var\big(\widehat{\beta}_{j,\text{OLS}}\big)$ is minimized when the covariates are orthogonal (as in Q.2).

4. Let the design matrix X has rank $p+1$. Then show that, for any parameter of the form $\underset{\sim}{c}^\top \beta$, the estimator $\underset{\sim}{c}^\top \widehat{\beta}_{\text{OLS}}$ is the best linear unbiased estimator.

5. Let

   $$Y_i = \beta_0 + \beta_1(x_{i1} - \bar{x}_1) + \beta_2(x_{i2} - \bar{x}_2) + \varepsilon_i, \qquad (i = 1, 2, \ldots, n),$$

   where

   $$\bar{x}_j = \sum_{i=1}^{n} \frac{x_{ij}}{n}, \qquad E[\varepsilon] = 0, \qquad \text{Var}[\varepsilon] = \sigma^2 I_n.$$

If $\hat{\beta}_1$ is the least squares estimate of β_1, show that

$$\mathrm{Var}[\hat{\beta}_1] = \frac{\sigma^2}{\sum_i (x_{i1} - \bar{x}_1)^2 (1 - r^2)},$$

where r is the correlation coefficient of the n pairs (x_{i1}, x_{i2}).

6. Let $Y_1, Y_2, \ldots, Y_n$ be n independent random variables with common variance σ^2 and common third and fourth moments, μ_3 and μ_4, respectively, about their means. If $E[\mathbf{Y}] = X\beta$, where X is an $n \times p$ matrix of rank p, then $(n-p)S^2$ is the unique nonnegative quadratic unbiased estimate of $(n-p)\sigma^2$ with minimum variance when $\mu_4 = 3\sigma^4$ or when the diagonal elements of P are all equal.

   [Hint: See Seber and Lee, Chapter 3, Section 3.3]

B. Problems with normality assumption on $\underset{\sim}{\varepsilon}$. Suppose in addition to the regression model in (1) we assume that $\varepsilon_i \overset{i.i.d.}{\sim} N(0,1)$.

1. Show that $\underset{\sim}{\widehat{\beta}}_{\mathrm{OLS}}$ and RSS are independent.

2. Show that both $\underset{\sim}{\widehat{\beta}}_{\mathrm{ML}}$ and $\widehat{\sigma}^2_{\mathrm{ML}}$ are consistent estimators.